

A domain-specific modal-growth detector clears a matched false-alarm bar on power-grid instability, and an eigenvalue regime map shows when its form transfers

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Abstract

Under a matched-false-alarm, permutation-tested evaluation protocol, generic early-warning detectors are statistically at chance on power-grid growing oscillations, as they are on brain, heart and climate data (companion manuscript). This paper shows the complementary positive result and bounds it. A domain-specific detector that reads the canonical wide-area-monitoring instability quantity — the exponential growth rate σ of the most unstable bus’s cross-bus voltage-deviation envelope — leads 36 of 90 generator-trip transitions of the PSML 23-bus corpus at a matched 10 % false alarm (label-permutation $p = 0.0001$; held-out half 24 of 45, $p = 0.0002$), on a non-circular disturbance-type split where the best generic detector is at chance (13 of 90, $p = 0.19$). The operating point was selected on a development half through a disclosed variant search and validated on the held-out half. Stress tests then bound the result. Run as a causal stream, the certified per-window operating point is not deployable, and an honest recalibration with a persistence debounce and an exponential-fit-quality gate recovers 11 of 45 held-out transitions at a matched 10 % stream false alarm. On two cross-dataset corpora (three real ISO-NE PMU captures; the 13 synthetic WECC 240-bus cases of the 2021 IEEE-NASPI contest), the certified numeric threshold does not transfer in either direction, while the frozen shape with per-system matched-false-alarm calibration remains usable. The form does not transfer to scalp EEG or to few-timepoint molecular indices. Finally, against ANDES small-signal eigenvalues of the IEEE-39 and Kundur systems, the analytic Jacobians of the fold, pitchfork and Hopf normal forms, and the mean-field eigenvalues of unimodal and bimodal Kuramoto models, the estimated quantity is confirmed to be the eigenvalue’s real part, yielding a regime map: the envelope-growth family sizes the eigenvalue on oscillatory modes, the autocorrelation family on non-oscillatory ones. All evidence records are content-addressed and byte-reproducible.

1 Background

When a disturbance leaves a power-system electromechanical mode under-damped, the amplitude of the cross-bus voltage deviation grows exponentially: the real part σ of the mode’s eigenvalue is positive, the canonical small-signal-stability quantity of wide-area monitoring (Kundur 1994). A detector that reads this quantity directly is domain-specific in the strict sense: it presupposes the physics of a linear instability rather than a generic statistical signature of critical slowing down.

A companion manuscript establishes the baseline this paper builds on: under a matched-false-alarm protocol with label-permutation significance — thresholds calibrated to a 10 % false

alarm on no-transition nulls, detection counted only when an alarm falls in the pre-onset segment, 10 000-draw permutation p-values, hash-sealed evidence — generic early-warning detectors (critical slowing down, rising synchronisation, ordinal-transition entropy, their fusion, and the Dakos AR(1)-Kendall- τ trend) are statistically at chance on grid data, as on brain, heart and palaeoclimate corpora. The questions here are: does a detector that reads the physical instability quantity clear that same bar; under what deployment conditions does the certified result survive; and is the quantity it estimates externally checkable?

2 Methods

2.1 The modal envelope-growth detector

The detector estimates σ directly as the slope of the log cross-bus voltage-deviation envelope over a scoring window. Two physically motivated refinements are selected on a development half only: a *focal* aggregation takes the growth rate of the most unstable bus (instability is localised to a mode/bus cluster, so the per-bus maximum, un-diluted, is the signal; the matched-false-alarm threshold is calibrated on that same per-bus maximum over the nulls, absorbing the multiplicity), and a *recency weighting* up-weights the later samples of the window (a real instability accelerates toward the disturbance). Every variant tried is disclosed in Section 3.2.

2.2 The evaluation protocol and the non-circular split

The protocol is the companion manuscript’s, unchanged: identical two-second pre-onset segments, thresholds calibrated continuously to a 10 % false-alarm target on the same nulls (achieved 9.1 %), a transition counted as led only on a pre-onset alarm, label-permutation significance with 10 000 draws and a fixed seed. The PSML corpus (Zheng et al. 2021; 23-bus co-simulation at 238 Hz) is split by disturbance *type*: a transition is any generator-trip scenario (90 usable), a null any damped bus-fault or branch-trip scenario (88) — a physical annotation independent of the growth statistic the detector measures, so a growing-oscillation detector is never scored against a growth-defined label. Three scenarios with a non-monotonic time column are dropped and the count is recorded in the sealed payload. The operating point is chosen on the even-index development half and validated on the odd-index held-out half.

2.3 A counterpoint detector on a domain without the signature

To test whether domain specificity alone wins, the same protocol runs a domain-specific detector on a domain without a deterministic signature: a scalp-EEG spectral detector (beta 13–30 Hz / delta 0.5–4 Hz band-power ratio, a-priori bands from the seizure-prediction literature, Kendall- τ rising trend, whole-head and focal aggregations) on the six chb01 preictal segments.

3 Results on the certification corpus

3.1 Head-to-head against the generic suite

The modal detector leads 36 of 90 transitions (40 %) at $p = 0.0001$; every generic member is at chance, the best (critical slowing down) reaching 13 of 90 at $p = 0.19$. The held-out half confirms the operating point: 24 of 45 (53 %) at $p = 0.0002$, the unbiased estimate.

Detector	Transitions led / 90	Permutation p
generic critical slowing down	13 / 90	0.185
generic weighted ensemble	11 / 90	0.324
generic ordinal-transition entropy	8 / 90	0.629
generic synchronisation	3 / 90	0.976
modal envelope-growth (focal, recency)	36 / 90	0.0001

Table 1: Identical segments, nulls, budget and test; only the detector differs.

3.2 The disclosed variant search

Variant	Development led / 45, p	Held-out led / 45, p
whole-network log-slope σ	11, 0.051	14, 0.009
per-bus maximum σ (focal)	14, 0.008	22, 0.0001
matrix-pencil, PCA-dominant mode	10, 0.072	3, 0.80
matrix-pencil, per-bus maximum	8, 0.189	6, 0.39
Hilbert-envelope σ , per-bus	2, 0.90	3, 0.80
recency-weighted focal σ	16, 0.002	24, 0.0002

Table 2: Choice made on development, reported on held-out; no held-out data touched during selection.

Notably, the gold-standard wide-area-monitoring estimator — matrix-pencil modal damping — recovers a planted σ exactly on synthetic data, yet on the short two-second, 238 Hz windows it is at chance (held-out $p = 0.39$ – 0.80); the robust direct log-envelope slope wins on real noisy segments. A recency-ratio sensitivity sweep (2–8) beats the unweighted per-bus detector across the whole range, so the choice is robust rather than knife-edge.

3.3 The counterpoint

The scalp-EEG spectral detector is itself at chance (1 of 6 led, $p \approx 0.56$, in both aggregations): the preictal state is murky and the corpus small. Domain specificity does not win by itself; the protocol certifies the difference between a detector reading a genuine physical signature and a plausible but empty one.

3.4 The streaming operating point

The head-to-head is a per-window result. Run as a live causal stream (every sliding window scored as a PMU feed would deliver it), the certified per-window threshold leads 82 of 90 transitions but false-alarms on 73 % of damped scenarios — a damped bus-fault or branch-trip produces a short transient growth window the continuous monitor alarms on, so the per-window operating point is operationally unusable as-is. Recalibrating at a matched stream-level false alarm, with a persistence debounce and an exponential-fit-quality gate that rejects a fault’s step-like transient (a fault fits an exponential poorly, a genuine instability well), the gated detector leads 11 of 45 held-out transitions (24 %) at a matched 10 % stream false alarm — well below the offline per-window 40 %. The gap is physical: separating a sustained instability from a damped fault online requires observing the damping sign over several cycles. The streaming operating point

— sealed in `examples/real_data/psml_modal_growth/grid_modal_stream_operating_point.json` — is the deployable figure, and the downstream advisory record is passive and review-only: it surfaces σ , the most-unstable bus and the certified operating point, carries the $\sim 24\%$ recall as a first-class field, never actuates, and states explicitly that the absence of an advisory is not evidence of stability.

4 Cross-dataset transfer

Two pre-registered cross-dataset evaluations — protocols fixed before any detector run, every amendment disclosed in the sealed records — test what survives a dataset change. Both are small, and the conclusions are scoped to these corpora.

The certified numeric threshold did not transfer on either corpus. On three real ISO-NE PMU captures of documented sustained oscillations (UTK test-cases library), the frozen PSML threshold ($1.3203\sigma/\text{s}$, two-second windows) fires on $\sim 32\%$ of pre-onset ambient windows against the certified 9.09% . On the 13 synthetic WECC 240-bus forced-oscillation cases of the 2021 IEEE-NASPI contest (bus-voltage observable, the certified observable family), the same threshold crosses nothing: 0 of 611 pooled ambient windows and 0 transition windows. On both corpora tested, the certified threshold behaved as a corpus-specific quantity; deployment therefore calibrates per system on the target system’s own ambient data and re-seals.

The frozen shape with local calibration, stated conservatively. On ISO-NE (window scaled to five cycles of the documented mode, threshold at a matched 10% false alarm on pooled pre-onset ambient windows), the detector catches 1 of 3 events, the 1.13 Hz regional mode, 57.1 s before the estimated onset ($p = 0.332$ at $n = 3$): a case study, no significance claim. On WECC the corpus structure dominates: 9 of 13 cases are already past three times the ambient in-band envelope at the exact forcing start ($t = 30\text{ s}$, known by simulation design) — an effectively instantaneous onset with no early-warning window for any detector at these 90 s records — and of the 4 resolvable cases the locally calibrated shape leads 2 ($+5.4\text{ s}$, $+4.1\text{ s}$; $p = 0.41$, not significant). A secondary branch reports post-onset detection descriptively (12 of 13, median latency $\sim 1\text{ s}$) with an explicit per-case chance bound showing the count is chance-compatible at this budget on 60 s regions; no significance is claimed. Records: `examples/real_data/iso_ne_forced_oscillation/` and `examples/real_data/wecc_240_osl/`.

5 Where the form does not transfer

Scalp EEG: a model mismatch. Scoring the same a-priori beta/delta band-power trajectory by the exponential growth rate of its envelope — identical focal aggregation, recency weighting and R^2 gate — leads 0 of 6 preictal segments ($p = 1.00$) in both aggregations, below the rank trend’s 1 of 6; the gate, built to reject non-exponential fault transients, rejects the preictal signal too. A robustness grid over window length, recency and gate stays at 0 of 6 at every setting: a model mismatch, not a parameter artefact — the statistic presupposes a linear instability’s eigenvalue, and the preictal rise, if present, is a slow non-exponential drift. (A 600 s horizon surfaces a rank-trend lead of 3 of 6, $p = 0.04$; one subject, one forking path — disclosed as exploratory, not claimed.) Record: `examples/real_data/chb01_seizures/seizure_modal_transfer.json`.

Few-timepoint molecular indices: a resolution limit. On the dynamical-network-biomarker rising limbs (three points single-cell, four points bulk), every monotone rise fits an exponential well enough ($R^2 = 0.67\text{--}0.86$) to pass the gate, and the growth rate re-orders the

lineages exactly as the existing linear slope does: the transfer collapses to the detector DNB already uses. Record: `examples/real_data/mojtahedi_fate/dnb_modal_transfer.json`.

Together these bound the transfer: the form needs both a genuine exponential instability and a trajectory long enough to fit and gate it.

6 External validation against eigenvalue ground truth

6.1 Real systems: ANDES small-signal analysis

On the IEEE 39-bus New England system and the Kundur two-area system (open-source ANDES simulator; Cui et al. 2021), we sweep the operating point, take the dominant electromechanical mode’s growth rate σ from small-signal eigenvalue analysis — an independent method — and score a ringdown from a small disturbance at each point. The *coherent* aggregation (cross-bus mean or dominant spatial mode) recovers the true σ trend on both systems (Spearman $\rho = 0.87$ on IEEE-39, 0.60–0.66 on Kundur); the *focal* aggregation — the PSML winner — does not ($\rho = -0.59, -0.27$): on a slow coherent inter-area mode the per-bus maximum locks onto local excursions. The quantity generalises; the best aggregation is regime-dependent. Limits: simulated ground truth, stable operating points only (a damping-ranking test), fourteen points per system. Record: `examples/real_data/grid_external_validation/grid_eigenvalue_external_validation.json`.

6.2 Analytic normal forms: fold, pitchfork, Hopf

For non-oscillatory codimension-one bifurcations the ground truth is closed-form: sweeping the control parameter quasi-statically through the stochastic normal form, the autocorrelation channel recovers the Jacobian eigenvalue λ ($-2\sqrt{\mu}$ fold, μ pitchfork) in rank and magnitude (Spearman and Pearson $\rho = 0.98/0.96$), because the lag-one autocorrelation of a linear-response process is $\exp(\lambda \Delta t)$; the variance channel rises in step ($\rho = 0.98$). On the Hopf normal form (oscillatory, eigenvalue $\alpha \pm i\omega$) the roles reverse: the envelope-growth family recovers α in rank and magnitude ($\rho = 0.97$, mean $|\sigma - \alpha| = 0.04$, frequency-invariant across 0.2–1.2 Hz), while the autocorrelation family tracks α only in rank (its lag-one autocorrelation is pinned near $\cos(\omega \Delta t)$). The envelope recovery degrades with ringdown SNR ($\rho 0.97 \rightarrow 0.62$), a physical floor. Records: `examples/real_data/csd_bifurcation/csd_bifurcation_external_validation.json`, `examples/real_data/hopf_bridge/hopf_bridge_external_validation.json`.

6.3 The collective coordinate: unimodal and bimodal Kuramoto

The same two regimes appear one level up, on the emergent collective coordinate of a high-dimensional system. For the noisy Kuramoto model ($N = 512$, Lorentzian frequencies, phase diffusion D), the incoherent state loses stability at $K_c = 2(\gamma + D)$ with a real mean-field eigenvalue $\lambda(K) = (K - K_c)/2$ (Sakaguchi 1988): the autocorrelation family recovers λ — but only through the signed mean-field component $\text{Re}(Z)$ (magnitude slope $1.19 \approx 1$); the folded amplitude $|Z|$ ranks it ($\rho = 0.98$) but cannot size it (slope 2.66). For a symmetric bimodal frequency law (Martens et al. 2009) the sub-critical eigenvalue is complex and the roles reverse exactly as the Hopf bridge predicts: the envelope-growth family on the sub-population order parameter $|Z_+|$ recovers $\text{Re}(\lambda)$ in magnitude ($\rho = 0.99$, slope 0.93; $N = 4000$ ringdown), the autocorrelation family is confounded by the oscillation Ω (slope 15.8), the global $|Z|$ is a standing wave that cannot be fit, and the measured Ω matches the analytic value to $\sim 7\%$. Limits: finite- N ($O(1/\sqrt{N})$), quasi-static or noiseless ringdown sweeps, coupling below K_c . Records: `examples/real_data/kuramoto_`

synchronization/kuramoto_synchronization_external_validation.json, examples/real_data/kuramoto_bimodal/kuramoto_bimodal_hopf_external_validation.json.

6.4 The regime map

Mode	Systems	Eigenvalue truth	Estimator	Recovery
Oscillatory, real (§6.1)	IEEE-39, Kundur	ANDES σ	envelope, coherent	rank (ρ 0.87 / 0.60+)
Non-oscillatory (§6.2)	fold, pitchfork	normal-form λ	AR1	magnitude (ρ 0.98 / 0.96)
Oscillatory (§6.2)	Hopf normal form	normal-form α	envelope σ	magnitude (ρ 0.97)
Collective, real λ (§6.3)	Kuramoto unimodal	mean-field λ	AR1 on $\text{Re}(Z)$	magnitude (slope 1.19)
Collective, complex λ (§6.3)	Kuramoto bimodal	mean-field $\text{Re}\lambda$	envelope on $ Z_+ $	magnitude (slope 0.93)

Table 3: The eigenvalue’s real part is the common quantity; the magnitude-correct estimator depends on whether the mode oscillates.

The split is mechanical, not empirical accident: on an oscillatory mode the lag-one autocorrelation is confounded by the oscillation while the envelope carries the decay cleanly; on a non-oscillatory mode there is no envelope to fit and the linear-response autocorrelation is exactly $\exp(\lambda \Delta t)$. Pick the estimator by whether the mode oscillates; the estimated quantity is the same eigenvalue throughout.

7 Discussion

The regime map of Table 3 is the paper’s main theoretical output. Across five independent ground truths — a power system simulator’s small-signal analysis, three analytic normal forms, and two mean-field Kuramoto reductions — the quantity the detector estimates is the eigenvalue’s real part, and which estimator recovers its magnitude is set mechanically by whether the mode oscillates: the envelope-growth family on oscillatory modes, the autocorrelation family on non-oscillatory ones, and on collective coordinates the signed mean field rather than the folded order-parameter amplitude. This turns the empirical grid result into a claim about a named first-principles quantity with a stated domain of applicability.

The empirical results bound that claim in deployment-relevant ways. A detector reading the instability quantity directly clears an operational bar that generic detectors do not, while the counterpoint shows domain specificity alone is not sufficient — the same idea on a domain without the signature stays at chance. The streaming recalibration replaces the per-window figure with an honest $\sim 24\%$ held-out recall at a held false alarm; the two cross-dataset corpora show the certified numeric threshold behaving as corpus-specific in both directions, leaving the frozen shape plus per-system calibration as the transferable unit; and the transfer boundaries on EEG and molecular data delimit where the exponential form applies at all.

8 Limitations

- The certification corpus is one dataset at one sampling rate with two-second windows; the held-out count (24/45) is the unbiased estimate and the full-corpus count (36/90) is the fitted-model deployment figure; both are sealed.
- The cross-dataset corpora are small (three real events; 13 synthetic cases, mostly with effectively instantaneous onsets), so the transfer conclusions are scoped to these corpora; the

WECC early-warning null partly reflects corpus structure rather than detector skill.

- The eigenvalue validations use simulated or analytic ground truths (no field-PMU eigenvalue reference), stable operating points or sub-critical sweeps, and finite- N Kuramoto runs.
- The EEG counterpoint and transfer test are one subject, six seizures.

9 Reproduction and data availability

Every result regenerates deterministically (fixed permutation seed; no other randomness). The pipelines are Python (NumPy/SciPy; pinned lock files in the repository); the PSML head-to-head and the external validations each complete in minutes on a workstation, the ANDES sweeps in tens of minutes. Entry points:

```
python -m bench.grid_modal_head_to_head          PSML OUT
python -m bench.early_warning_leadtime_isone     --data-dir ISONE --output OUT
python -m bench.early_warning_leadtime_wecc      --data-dir WECC  --output OUT
python -m bench.csd_bifurcation_external_validation  OUT
python -m bench.dnb_modal_transfer                OUT
```

Sealed records and integrity tests are committed under `examples/real_data/` in the source repository (<https://github.com/anulum/scpn-phase-orchestrator>); a fresh run reproduces every `content_hash` bit-for-bit. Raw corpora are public and cited, not redistributed: PSML (Zheng et al. 2021), the ISO-NE and WECC 240-bus cases of the UTK oscillation test-cases library (<https://web.eecs.utk.edu/~kaisun/Oscillation/>; Maslennikov et al. 2016; NREL reduced 240-bus model, credit DOE/NREL/Alliance), and CHB-MIT via PhysioNet.

Declaration of AI use

AI-assisted technologies (Anthropic Claude large-language-model tooling) were used to assist with software development, analysis tooling and manuscript drafting. The author specified, verified and approved all methods, analyses and claims, and takes full responsibility for the content of this work.

Competing interests

The author develops the open-source SCPN Phase Orchestrator software used in this study, which is also offered under a commercial licence.

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